

Course Code	21ECC302T	Course Name	ANALOG AND DIGITAL COMMUNICATION	Course Category	C	PROFESSIONAL CORE	L	T	P	C
							3	0	0	3

Pre-requisite Courses	21MAB203T	Co-requisite Courses	Nil	Progressive Courses	Nil
Course Offering Department	ECE	Data Book / Codes / Standards	Nil		

Course Learning Rationale (CLR):		Program Outcomes (PO)												Program Specific Outcomes		
The purpose of learning this course is to:		1	2	3	4	5	6	7	8	9	10	11	12	PSO-1	PSO-2	PSO-3
CLR-1:	introduce to the learners the basic concepts involved in Communication system	Engineering Knowledge	Problem Analysis	Design/development of solutions	Conduct investigations of complex problems	Modern Tool Usage	The engineer and society	Environment & Sustainability	Ethics	Individual & Team Work	Communication	Project Mgt. & Finance	Life Long Learning			
CLR-2:	comprehend the functionalities of various radio transmitters and receivers															
CLR-3:	realize the process involved in digital communication systems															
CLR-4:	explore the pass band transmission system and analyze its performance in terms of probability of error															
CLR-5:	get exposed to Information theory and channel coding concepts															
Course Outcomes (CO):		At the end of this course, learners will be able to:														
CO-1:	explain the various analog modulation techniques	3	-	-	-	-	-	-	-	-	-	-	2	2	-	-
CO-2:	analyze the noise performance of radio transmitters and receivers	3	3	-	-	-	-	-	-	-	-	-	2	-	3	-
CO-3:	demonstrate the demodulation and detection of received digital data	3	2	-	-	-	-	-	-	-	-	-	-	-	-	3
CO-4:	apply the suitable passband techniques for real time applications	3	-	-	-	3	-	-	-	-	-	-	-	-	-	2
CO-5:	exposed to the concepts of information theory and channel capacity	3	-	3	-	-	-	-	-	-	-	-	-	3	-	-

Unit-1 - Analog Modulation Techniques	9 Hour
Need for Modulation - Types of Analog Modulation - Amplitude Modulation (AM) and its types - Generation of AM Waves - Linear Method (Collector Modulator) - Non Linear Method (Balanced Modulator) - Demodulation of AM waves (Envelop Detector) - Frequency Modulation (FM) – Types of FM –Narrow Band FM (NBFM) and Wide Band FM (WBFM) - Generation of NBFM (Varactor Diode Modulator) - Demodulation of NBFM waves (Foster Seely Method)- Phase Modulation (PM)- Generation of PM from FM and FM to PM	
Unit-2 - Radio Transmitters and Receivers	9 Hour
AM Transmitter (Low Level and High Level) - FM Transmitter (Direct and Indirect Method) - Characteristics and functions of a receiver - AM Superheterodyne Receiver and FM Super Heterodyne Receiver - Noise in AM and FM (Elementary Treatment) - Need for Pre-emphasis and De-emphasis circuits	
Unit-3 - Baseband and Digital Modulation Techniques	9Hour
Baseband Modulation Techniques (PAM, PWM and PPM) - Digital Modulation Techniques - Pulse Code Modulation (PCM) System) - Differential PCM (DPCM) System - Delta Modulation (DM) System - Matched Filter Receiver - Probability of error for Matched filter - Inter Symbol Interference (ISI) and Eye pattern	
Unit-4 - Passband Transmission System	9 Hour
Passband Transmission System Model – Passband Modulation Techniques- Generation, Signal Space diagram, Detection, Probability of Error for BFSK - BPSK – QPSK – M-ary PSK and FSK (Elementary Treatment) – QAM System	
Unit-5 - Information Theory and Channel Capacity	9 Hour
Entropy, Information rate, Source coding theorem, Shannon-Fano coding, Huffman coding, Mutual information – Shannon's channel capacity theorem	

Learning Resources	1. Simon Haykin and Michael Moher, "Communication Systems," 5th edition, John Wiley & Sons, 2013	4. Bernard Sklar, "Digital Communication, Fundamentals and Application", Pearson Education Asia, 2nd Edition, 2001
	2. Singh. R. P & Sapre. S. D, "Communication Systems: Analog & Digital," 3rd edition, Mc GrawHill Education, Seventh Reprint, 2016.	5. Taub & Schilling, "Principle of Communication Systems", McGraw Hill Inc, 2nd Edition, 2003.
	3. Simon Haykin, "Communication Systems", John Wiley & Sons, 4th Edition, 2008	6. John G. Proakis, "Digital Communication", McGraw Hill Inc, 5th Edition, 2008.

Learning Assessment							
	Bloom's Level of Thinking	Continuous Learning Assessment (CLA)				Summative Final Examination (40% weightage)	
		Formative CLA-1 Average of unit test (50%)		Life-Long Learning CLA-2 (10%)			
		Theory	Practice	Theory	Practice	Theory	Practice
Level 1	Remember	25%	-	15%	-	15%	-
Level 2	Understand	25%	-	20%	-	25%	-
Level 3	Apply	30%	-	25%	-	30%	-
Level 4	Analyze	20%	-	25%	-	30%	-
Level 5	Evaluate	-	-	10%	-	-	-
Level 6	Create	-	-	5%	-	-	-
	Total	100 %		100 %		100 %	

Course Designers		
Experts from Industry	Experts from Higher Technical Institutions	Internal Experts
1. Mr. Anuj Kumar, Bombardier Transportation, Ahmedabad, kumaranuj.anii@gmail.com	1. Dr. Meenakshi, Professor of ECE, CEG, Anna University, meena68@annauniv.edu	1. Dr. M. Sangeetha, SRMIST
2. Mr. Hariharasudhan - Johnson Controls, Pune, hariharasudhan.v@jci.com	2. Dr. Venkatesan, Sr. Scientist, NIOT, Chennai, venkat@niot.res.in	